INCH-POUND

MIL-P-63223A (AR)
AMENDMENT 3
03 August 1994
SUPERSEDING
AMENDMENT 2
12 July 1989

MILITARY SPECIFICATION

PROJECTILE, GUIDED, 155MM, M712

This Amendment forms a part of Military Specification MIL-P-63223A (AR), dated 5 December 1984, and is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 7

4.3.2, Insert the following between the word "drawings" and the word "Environmental" in line 4:

"The scoring criteria shall be in accordance with the reliability scoring criteria in the Appendix."

4.3.3, Delete the first five (5) lines and substitute the following:

"4.3.3 Rejection. If any test sample fails to comply with any of the applicable requirements, the first article samples shall be rejected. When launched, not more than four (4) projectiles in each temperature group of 10 projectiles and not more than seven (7) projectiles totally shall fail to function as defined in 3.3.1.3 or the first article sample shall be rejected."

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* 4.4.2.5 Major Category change:

From: "109. Painting of dome (window)"

To: "109. Painting of dome (window) workmanship and dimensions"

Minor 206 delete in its entirety and substitute the following:

"206 Preparation for delivery 100% 5.0 4.5.8"

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4.4.3.1 Delete the entire paragraph and substitute the following:

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"4.4.3.1 Sequential sampling for flight testing. Each inspection lot shall be subjected to the flight test by the procuring activity in accordance with 4.5.7. The scoring criteria shall be in accordance with the reliability scoring criteria in the Appendix. Sampled projectiles from each inspection lot shall be selected at random by the procuring activity for the flight test. The test samples shall be selected from the production run throughout the period that an inspection lot is being produced. After selection, the test sample shall be stored in containers (MIL-P-63227) in a bonded area and no further tests on the sample shall be performed by the contractor. The acceptance/rejection criteria shall be in accordance with the sequential sample plan of Table V if there are two or less soft recovery failure, and the criteria shall be in accordance with the sequential sample plan of Table VI if there are three or more soft recovery failures. If the number of failures for the six soft recovery rounds is equal to 2, and the score for the six-soft recovery rounds in the previous lot was 2, the sample size and criteria for lot acceptance test shall be as shown in Table VI. If the failure mode(s) indentified in a failed soft recovery round is such that the failure mode(s) can be eliminated from the lot by screening or rework, the Contractor may screen or rework the lot to remove the rounds with the failure mode(s). In this event, the failed soft recovery round(s) shall not be scored as a failure. The Government reserves the right to implement the sample size and acceptance criteria as shown in Table V without regard to the scoring results of the soft recovery tests. In event that an accept/reject decision is determined before all selected projectiles have been flight tested, the remaining projectiles shall be returned to the lot for delivery. The environmental test performed on the projectile is not considered destructive and projectiles that have been subjected to environmental tests and not flight tested shall be returned to the lot for delivery. The cannon launch test conditions shall be in accordance with Table IV."

4.4.3.2, Delete the word "three (3)" on line 2 of the first paragraph and substitute the word "six (6)".

4.4.3.2, Delete the first sentence in the last paragraph and substitute the following:

"The contractor shall select three (3) projectiles from the beginning of the month and three (3) projectiles from the middle of the month. In each lot, at least two of the soft recovery rounds shall be separated at the seeker electronics package interface and at least two (2) rounds shall be seperated at the guidance section/warhead interface. Three (3) rounds, shall be

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tested at cold temperature (-25 degrees F), three (3) rounds shall be tested at high temperature (+145 degrees F) and all six (6) rounds shall be fired at high g using the M119A2 zone 8 charge. In all cases, the rounds shall be preconditioned to the same temperature for the live sequence phase of the test as they were during the gunfire phase of the test.

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Delete Table V and substitute Tables V and VI as follows:
(Note: Table V was previously changed by Amendment 1.)

"Table V. Sequential sample plan and accept/reject criteria.

Number samples	Number Failures	
	Accept	Reject
1	-	-
2	-	-
3	-	3
4	-	4
5	0	4
6	0	4
7	0	4
8	0	5
9	1	5
10	1	5
11	1	5
12	1	5
13	2	6
14	2	6
15	2	6
16	2	6
17	3	7
18	3	7
19	3	7
20	3	7
21	4	8
22	4	8
23	4	8
24	4	8
25	5	9
26	5	9
27	5	9
28	5	9
29	6	10
30	6	10
31	7	10
32	8	10
33	9	10

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NOTES: (a) The number of failures stated above represents functional failures as defined by paragraph 4.5.1.4. A failure of this type is classified as a Major Defect.

(b) If a projectile detonates prior to fuze arming the failure is classified as a Critical Defect and the entire lot is rejected.

Table VI. Sequential sample plan and accept/reject criteria.

Number Samples	Number Failures	
	Accept	Reject
1	-	-
2	-	-
3	-	-
4	-	4
5	-	4
6	-	5
7	0	5
8	0	5
9	0	5
10	0	5
11	0	6
12	1	6
13	1	6
14	1	6
15	1	6
16	1	7
17	2	7
18	2	7
19	2	7
20	2	7
21	2	8
22	3	8
23	3	8
24	3	8
25	3	8
26	3	9
27	4	9
28	4	9
29	4	9
30	4	9
31	5	10
32	5	10
33	5	10
34	5	10
35	5	11
36	6	11
37	6	11

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38	6	11
39	6	11
40	6	11
41	7	11
42	7	11
43	7	11
44	7	11
45	7	11
46	8	11
47	9	11
48	10	11

NOTES: (a) The number of failures stated above represents functional failures as defined by paragraph 4.5.1.4. A failure of this type is classified as a Major Defect.

(b) If a projectile detonates prior to fuze arming, the failure is classified as a Critical Defect and the entire lot is rejected."

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4.5.8, Delete line 4 and substitute the following:

"been prepared and soft recovered per 4.5.8.1 and figures 7 and 8. The projectile"

4.5.8, Add figures 7 and 8 included.

PAGE 30

Add the following paragragh:

4.5.8.3 Soft recovery scoring criteria

1. Projectiles utilized for soft recovery firing in accordance with 4.5.8.1 shall be subject to the following scoring criteria:

a. During gunfire testing, rounds which experience hard impact due to failure of the parachute recovery system will be scored as "NO TEST".

b. During continuity testing at the firing site, failures will be scored only for opens or shorts that are verified by subsequent laboratory testing.

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CONTRACTOR OPERATION

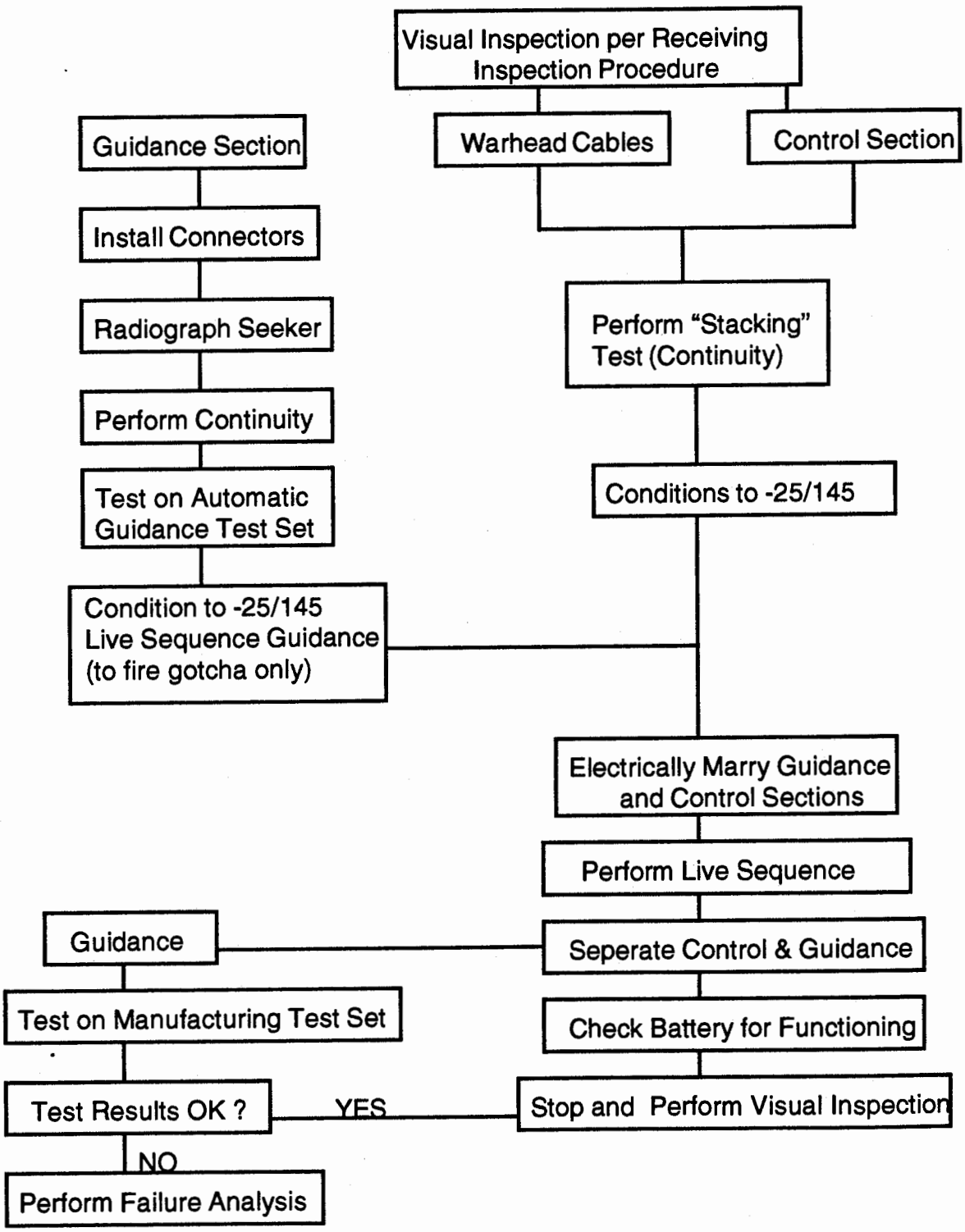


Figure 7

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WSMR OPERATION

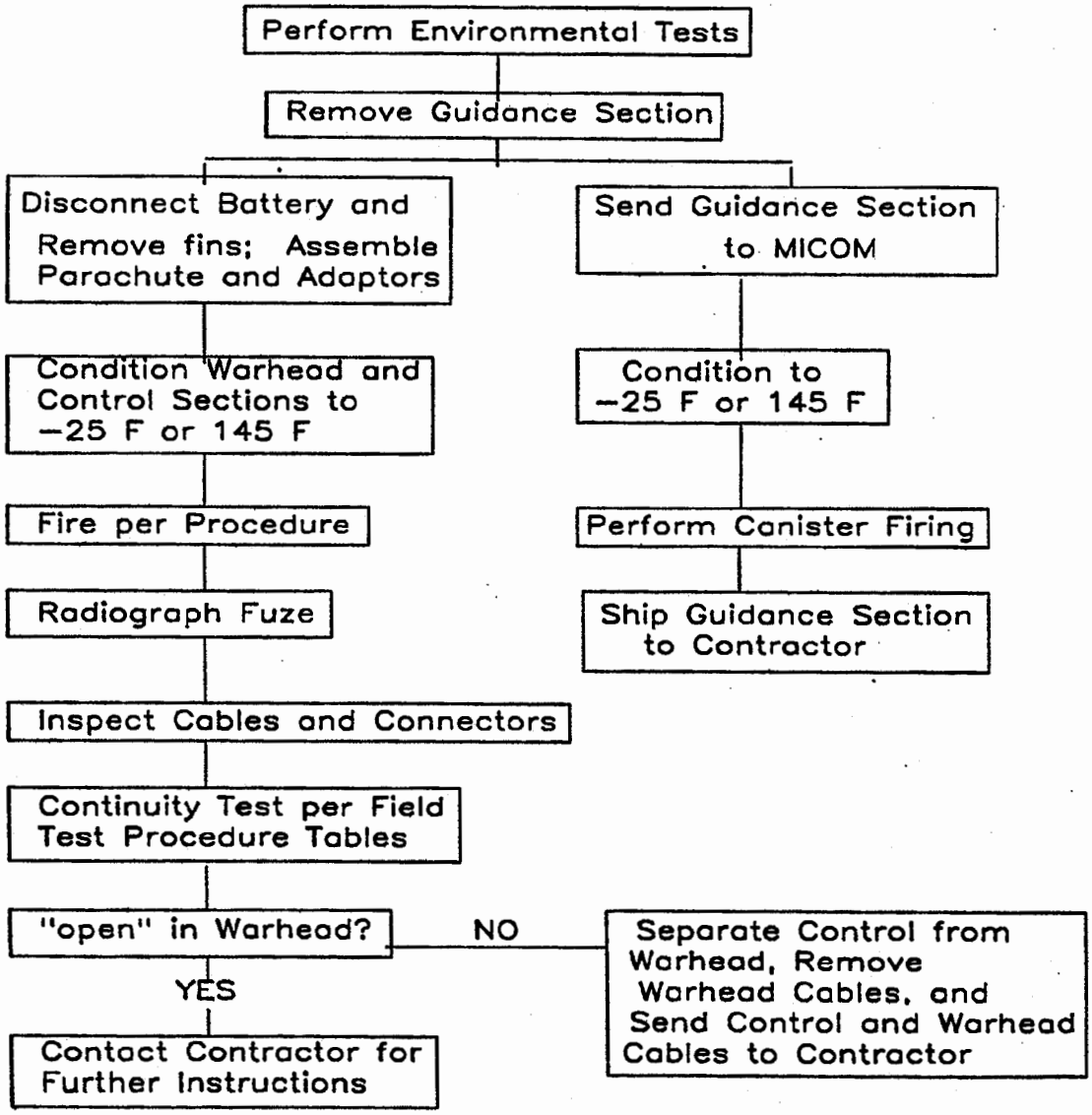


Figure 8

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2. Projectiles utilized for live sequence testing in accordance with 4.5.8.2 shall be subject to the following scoring criteria:

a. A failure shall be scored for any confirmed failure which would result in a flight failure

b. Scoring will not be based on tolerance measurements made from the live sequence records.

c. Unverified anomalies will not be scored as failures.

d. Intermittents caused by dirt in the control actuation system will not be scored as a failure.

e. If a round is scored as a failure due to the failure of the gyro gotcha to drop and if this scored failure causes the subsequent LAT to be conducted in accordance with the accept reject criteria of Table VI, then the lot shall be accepted if the firt five (5) rounds are successful.

f. During guidance section ATE testing, the following shall apply:

1) A failure due to the roll rate sensor shall be scored only for catastrophic failure of the roll rate sensor.

2) A failure shall be scored due to parameter values when the parameter is outside allowable tolerances and deviates more than ten (10) percent from the previously measured value and is not due to test equipment malfunction.

3) The occurrence of a problem that would not result in a flight failure will be scored as a success unless the problem occurs more than twice.

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Add Appendix included

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APPENDIX

RELIABILITY SCORING CRITERIA
FOR
M712 COPPERHEAD (CLGP)
CANNON LAUNCHED GUIDED PROJECTILE

10. SCOPE.

10.1 Scope. This scoring criteria establishes guidelines for failure determinations on the M712 Copperhead Cannon Launched Guided Projectile (CLGP). Failure types and reliability impact assessment criteria are defined. Paragraph 30 is the Copperhead Failure Definition and forms the basis for the Scoring Procedures of Paragraph 50. The Operational Mode Summary in Paragraph 40 outlines the basic use of Copperhead and the range of conditions over which test data will be scored. The Decision Table Flow Chart in Paragraph 60 provides a summary diagram of the scoring procedures detailed in Paragraph 50.

20. APPLICABLE DOCUMENTS.

20.1 Government documents.

20.1.1 Specifications and standards. Unless otherwise specified the following specifications and standards listed in that issue of the Department of Defense Index of Specifications (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-882 - System Safety Program Requirements

30. FAILURE DEFINITION.

30.1 Requirements. The procedure of scoring has been structured to identify test data suitable for use in assessments of reliability.

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30.2 General failure definition. For the purpose of assessing reliability, a Copperhead failure is defined as any malfunction which causes or could cause inability of the projectile to perform its intended function or a critical or catastrophic hazard to personnel or equipment. When more than one malfunction occurs in a test, only the first malfunction will be counted against reliability.

30.3 Clarification of terms.

30.3.1 Intended function. This term means the projectile must successfully survive prelaunch environment and handling, survive launch shock, and perform properly the actions to power-up, stabilize, shape trajectory (if required), acquire, correlate, arm, manuver to, and detonate at the target.

30.3.2 Critical or catastrophic hazards. These terms are defined by MIL-STD-882. An example of a failure to this classification would be a partially armed fuze prior to firing. While the projectile might still function correctly, the hazard to personnel and equipment is unacceptable.

30.4 Types of failures not chargeable against projectile reliability.

a. A failure to function caused by a malfunction of the interfacing equipment or systems, e.g., power failure of the designator, or incorrect arming solution or procedure.

b. A malfunction, real or potential, resulting from not following the prescribed operational or maintenance procedure in the M712 Technical Manuals.

c. A malfunction resulting from accidental test item abuse or testing beyond specification limits and conditions.

d. Any secondary or follow-on malfunction when more than one malfunction occurs during a test.

30.5 Types of failure, chargeable against projectile reliability.

a. A malfunction which would prevent the projectile from performing its intended function.

b. A malfunction which would cause a critical or catastrophic hazard to personnel or equipment.

c. The first or initial malfunction when more than one malfunction occurs.

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d. A malfunction or problem detected prior to launch which could cause the projectile to malfunction or present a hazard.

e. A malfunction, real or potential, caused by following the prescribed operational or maintenance procedures in the M712 Technical Manuals.

40. M712 OPERATIONAL MODE SUMMARY

a. The M712 Cannon Launched Guided Projectile (Copperhead) will be used in indirect fire by 155mm field artillery units to destroy or neutralize moving and stationary point targets, such as armored and mechanized vehicles, field fortification, small logistic points, small groups of personnel, and command or observation posts.

b. The operational Copperhead can be expected to undergo shipping, storage, deployment, transportation, handling, loading, firing, flight to and detonation at a laser designated target.

c. The Copperhead System Specification, ARC 100, dated 12 July 1978, contains the performance requirements that Copperhead must meet. The following set of stated conditions have been established as the range of conditions from which test data will be scored for reliability:

40.3.1 Climate: Operating categories 1-6, storage categories 1-7, AR 70-38.

40.3.2 Cloud cover: Not below 1,000 feet AGL.

40.3.3 Wind speed: 50 knots (any direction) between 0 and highest par of trajectory.

40.3.4 Gun to target range: Maximum - 16,000
Minimum - 5,000 meters (high angle)
3,000 meters (low angle)

40.4.5 Maximum designator to target range

Stationary Targets 6,000 meters

40.3.6 Gun target - observer target angle

Between + or - 45 degrees from firing azimuth

40.3.7 Visibility range: 7,500 meters

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40.3.8 Target type: 16 ft. x 16 ft. stationary flat black panel

40.3.9 Designator performance: GLLD specifications or equivalent.

40.3.10 Target: Reflectivity is equal to or greater than 10%.

50. SCORING PROCEDURE.

50.1 General information. So the scoring committee can better use the procedures defined below, the following information and definitions are provided.

50.2 Projectile configuration. This scoring criteria has been structured to evaluate fully operational projectiles.

50.3 Repaired projectiles. Any projectile which after acceptance by the Government exhibits a malfunction and is returned to the contractor, repaired and resubmitted for testing will be treated as a new projectile. This could mean a projectile would be scored more than once.

50.4 Potential malfunction. Any projectile component, parameter, or characteristic found, by inspection, to be in a state that would cause the projectile to fail to function reliably, will be scored a projectile failure. The disposition of the projectile will be the responsibility of the PMO and could include return for repair, continue with restricted testing, or continue regular testing. If the projectile testing is continued and is against any performance requirement in the system specification, the projectile's success or failure to function reliably will be scored.

50.5 Target engagement analysis. In assessing projectile reliability it is important to know if an incident where the projectile failed to impact the target area was attributed to a malfunction of the projectile or a set of conditions which placed the target location outside the engagement (performance) envelope of the projectile. For example, suppose data from two different tests indicated that both projectiles acquired the targets and maneuvered in the direction of the targets; however, both impacted outside of the required distance in order to be scored a reliability success. An analysis of target location with respect to the projectile engagement capability (perhaps a visibility and maneuverability overlay for a given test) indicated that, in one case, the target's location was such that the projectile could not have successfully engaged it; and, in the second case, the projectile should have engaged the target. The first case data would be classed as a no test from the standpoint of reliability.

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Target engagement analyses will be required whenever conditions are suspected of contributing to the poor performance of the projectile. If it is found that the target is outside of or perhaps at the very edge of projectile performance, these data should not be scored for reliability. This may have the effect of throwing our test successes under the same conditions. However, those could be the exception to nominal performance and should, therefore, not be included in the analysis of reliability.

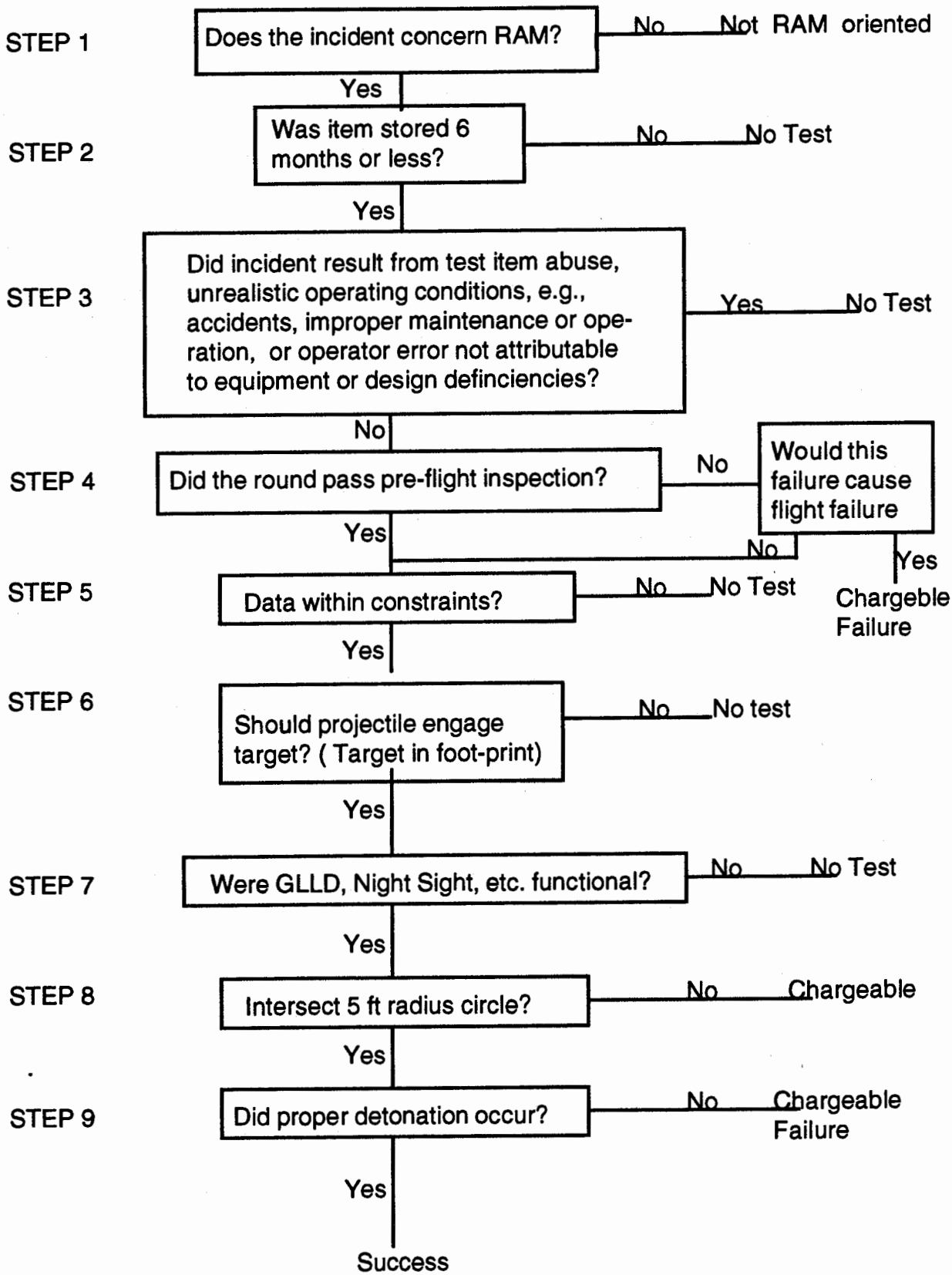
50.6 Designator spot centroid. The centroid of laser spot energy averaged over the last three seconds of projectile flight prior to intersection with a plane perpendicular to the velocity vector and through the averaged centroid shall define the plane in which the 5-foot radius circle will exist.

60. Decision table flow chart. The flow chart is shown on the following page."

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DECISION TABLE FLOW CHART.



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